

Logic in AI : Resolution - Part - 5

Introduction

In the realm of Artificial Intelligence, logic serves as a fundamental language for communication between humans and AI agents. It provides a structured way to represent knowledge about the world, enabling machines to perform logical inferences and respond to queries based on their current understanding. This logical foundation allows AI systems to reason, deduce new facts, and make informed decisions within a given domain.

Previous discussions explored the basics of logical representation, including syntax and semantics, and introduced inference algorithms like forward chaining. Forward chaining, typically used with Horn clauses and Modus Ponens, proves new facts by moving from known premises to conclusions. While sound, it is only complete for specific scenarios, such as when the knowledge base consists solely of Horn clauses and queries are single literals or conjunctions of literals. Backward chaining, another inference method, works in reverse, starting from a desired conclusion and working backward to find supporting evidence in the knowledge base.

This lecture introduces resolution proof systems, a powerful and widely applicable inference technique. Resolution is equivalent to proof by contradiction and requires all logical formulas to be converted into a canonical form known as Conjunctive Normal Form (CNF). This standardization simplifies the inference process, allowing for a consistent approach to logical reasoning in AI systems.

Core Concepts

Term	Definition	Significance
Conjunctive Normal Form (CNF)	A standardized logical formula represented as a conjunction of disjunctions, where each disjunction is called a clause.	It provides a canonical representation for any logical formula, simplifying the application of inference rules like resolution.
Clause	A disjunction of literals (e.g., A or not B or C).	Clauses are the fundamental building blocks of CNF formulas, which are essential for resolution-based inference.

Term	Definition	Significance
Resolution Rule	An inference rule that, given two clauses containing complementary literals (e.g., P and $\neg P$), produces a new clause containing all other literals from the original clauses.	It is a sound inference rule that forms the basis of resolution proof systems, enabling the derivation of new logical consequences.
Soundness	A property of an inference algorithm where any conclusion derived from the premises is logically entailed by those premises.	Ensures that the resolution rule only produces valid conclusions, preventing the derivation of false statements from true ones.
Refutation Completeness	A property of an inference system meaning that if a set of clauses is unsatisfiable (i.e., no model can make them all true), the system can always derive an empty clause (representing false).	While not complete in proving all true statements, it guarantees that contradictions can always be detected, which is crucial for proof by contradiction.
Proof by Contradiction	A proof technique where one assumes the negation of what is to be proven and then demonstrates that this assumption leads to a logical contradiction (an empty clause).	This is the primary method used with resolution to prove a statement; if negating the goal leads to a contradiction, the original goal must be true.
Empty Clause	A clause containing no literals, which is equivalent to logical falsehood.	Deriving an empty clause signifies a contradiction, indicating that the initial set of clauses (including the negated goal) is unsatisfiable.
Modus Ponens	A basic inference rule stating that if " P implies Q " is true and	Resolution is shown to be more powerful than Modus

Term	Definition	Significance
	"P" is true, then "Q" must be true.	Ponens, as it can derive conclusions that Modus Ponens can, and more.

Detailed Analysis

1. Converting to Conjunctive Normal Form (CNF)

The first crucial step in using resolution is to convert all logical formulas into Conjunctive Normal Form (CNF). This standardization ensures that the resolution rule can be applied uniformly. The conversion process follows a specific order of operations:

- **Eliminate Bidirectional Implications:** Any formula of the form $A \text{ if and only if } B$ ($A \iff B$) is converted into $(A \text{ implies } B) \text{ AND } (B \text{ implies } A)$. This replaces one bidirectional implication with two unidirectional implications and a conjunction.
 - **Convert Unidirectional Implications to Disjunctions:** Every implication $\alpha \text{ implies } \beta$ ($\alpha \implies \beta$) is transformed into $\text{NOT } \alpha \text{ OR } \beta$. This step ensures that all implications are expressed using only disjunctions and negations.
 - **Apply De Morgan's Laws:** If negations appear outside parentheses containing conjunctions or disjunctions (e.g., $\text{NOT } (A \text{ AND } B)$ or $\text{NOT } (A \text{ OR } B)$), De Morgan's laws are applied to push the negations inward.
 - $\text{NOT } (A \text{ AND } B)$ becomes $(\text{NOT } A \text{ OR } \text{NOT } B)$.
 - $\text{NOT } (A \text{ OR } B)$ becomes $(\text{NOT } A \text{ AND } \text{NOT } B)$.
 - **Apply Distributive Laws:** If a disjunction is distributed over a conjunction (e.g., $A \text{ OR } (B \text{ AND } C)$), the distributive law is used to transform it into a conjunction of disjunctions.
 - $A \text{ OR } (B \text{ AND } C)$ becomes $(A \text{ OR } B) \text{ AND } (A \text{ OR } C)$.
- This step is repeated until the entire formula is a conjunction of clauses, where each clause is a disjunction of literals. Once in CNF, the formula is ready for resolution.

2. The Resolution Rule and its Properties

The core of the resolution proof system is the resolution rule. This rule operates on two clauses to produce a new clause, known as the resolvent.

- **Basic Resolution Rule:** If you have two clauses, $(P \vee \alpha)$ and $(\neg P \vee \beta \vee \gamma)$, where P is a literal and $\neg P$ is its complement, the resolution rule allows you to infer $(\alpha \vee \beta \vee \gamma)$. The complementary literals P and $\neg P$ are "resolved away."
- **Soundness:** The resolution rule is sound. This means that if the premises (the original clauses) are true, then the conclusion (the resolvent) must also be true. In other words, if a set of clauses S_1 can be used to derive S_2 via resolution, then S_1 logically entails S_2 .
- **Refutation Completeness:** Resolution is refutation complete. This is a powerful property, meaning that if a set of clauses is unsatisfiable (i.e., there is no possible assignment of truth values that makes all clauses true simultaneously), then resolution can always derive an empty clause (which represents logical falsehood). This is crucial for proof by contradiction. It's important to note that resolution is not "complete complete" in the sense of Gödel's incompleteness theorems, which state that no sufficiently expressive proof system can prove all true statements. However, for the purpose of detecting contradictions, refutation completeness is sufficient.

3. Resolution's Relationship with Modus Ponens

Resolution is a more general and powerful inference rule than Modus Ponens. It can effectively perform any inference that Modus Ponens can.

- **Example: Modus Ponens via Resolution:**
Consider the Modus Ponens inference: If $A \text{ implies } B$ and A is true, then B is true.
 - Convert $A \text{ implies } B$ to CNF: $\neg A \vee B$.
 - The premise A is already a clause: A .
 - Apply resolution to $(\neg A \vee B)$ and (A) . The complementary literals $\neg A$ and A resolve away.
 - The resolvent is B .This demonstrates that resolution can derive B from $A \text{ implies } B$ and A , just as Modus Ponens would. This shows resolution's strength and generality.

4. Proof by Contradiction using Resolution

The primary method for proving a statement using resolution is proof by contradiction.

- **Steps for Proof by Contradiction:**
- **Convert Knowledge Base to CNF:** All known facts and rules in the knowledge base are converted into a set of CNF clauses.
- **Negate the Goal:** The statement you want to prove (the query) is negated.
- **Convert Negated Goal to CNF:** The negated goal is also converted into CNF clauses.
- **Add to Knowledge Base:** The CNF clauses of the negated goal are added to the existing set of clauses from the knowledge base.
- **Apply Resolution Repeatedly:** The resolution rule is applied repeatedly to pairs of clauses in the combined set. Each time a new clause is generated, it is added to the set.
- **Derive Empty Clause:** If an empty clause (representing `false`) is derived, it means that the combined set of clauses (knowledge base + negated goal) is unsatisfiable. Since the knowledge base is assumed to be consistent, the contradiction must arise from the negated goal. Therefore, the original goal must be true.
- **Failure to Derive Empty Clause:** If resolution cannot derive an empty clause after all possible resolutions have been performed, it means the negated goal is consistent with the knowledge base, and thus the original goal cannot be proven true by this method.

5. Example Walkthrough: The Unicorn Problem

Let's illustrate proof by contradiction with resolution using the "Unicorn Problem":

- * "If the unicorn is mythical, then it is immortal."
- * "If the unicorn is not mythical, it is a mammal."
- * "If the unicorn is either immortal or a mammal, then it is horned."
- * **Goal:** "The unicorn is horned."

Symbol Definitions:

- * M: Mythical
- * I: Immortal
- * A: Mammal
- * H: Horned

Conversion to CNF Clauses:

1. "M implies I" -> `NOT M OR I`

2. "NOT M implies A" $\rightarrow M \vee A$

3. "(I OR A) implies H" $\rightarrow \text{NOT } (I \vee A) \vee H \rightarrow (\text{NOT } I \wedge \text{NOT } A) \vee H \rightarrow (\text{NOT } I \vee H) \wedge (\text{NOT } A \vee H)$

So, this gives two clauses: $\text{NOT } I \vee H$ and $\text{NOT } A \vee H$.

Negate the Goal:

* Goal: H

* Negated Goal: $\text{NOT } H$

Resolution Steps (to derive empty clause):

1. Start with the negated goal: $\text{NOT } H$

2. Resolve $\text{NOT } H$ with $\text{NOT } I \vee H \rightarrow \text{NOT } I$ (resolving H and NOT H)

3. Resolve $\text{NOT } H$ with $\text{NOT } A \vee H \rightarrow \text{NOT } A$ (resolving H and NOT H)

4. Resolve $M \vee A$ with $\text{NOT } A \rightarrow M$ (resolving A and NOT A)

5. Resolve $\text{NOT } M \vee I$ with $\text{NOT } I \rightarrow \text{NOT } M$ (resolving I and NOT I)

6. Resolve M with $\text{NOT } M \rightarrow \text{EMPTY CLAUSE}$

Since an empty clause is derived, the initial assumption ($\text{NOT } H$) leads to a contradiction. Therefore, the original goal, "The unicorn is horned," must be true.

6. Implementing Resolution Systems

Implementing a resolution proof system involves a search process:

- **Database of Clauses:** Maintain a database of all CNF clauses, including those from the knowledge base and the negated goal.
- **Looping for Resolution:** Continuously select two clauses from the database, apply the resolution rule, and add the newly generated resolvent to the database.
- **Control Strategies:** To manage the potentially vast search space, control strategies (heuristics) are used to decide which pair of clauses to resolve next.
- **Success Condition:** If an empty clause is generated, the proof is successful.
- **Failure Condition:** If no new clauses can be generated, and an empty clause has not been found, then the proof fails (meaning the goal cannot be proven by contradiction).
- **Optimization:** To prevent redundant work, ensure that no pair of clauses is resolved more than once. Also, simplify clauses by removing duplicate literals (e.g., $Q \vee P \vee P$ becomes $Q \vee P$).

Key Takeaways

- [] Logic is a primary means of communication and knowledge representation for AI agents.
- [] Resolution is a powerful inference algorithm equivalent to proof by contradiction.
- [] All logical formulas must be converted to Conjunctive Normal Form (CNF) before applying resolution.
- [] The CNF conversion process involves eliminating bidirectional implications, converting unidirectional implications, applying De Morgan's laws, and applying distributive laws.
- [] The resolution rule combines two clauses with complementary literals to produce a new clause.
- [] Resolution is a sound inference rule, ensuring derived conclusions are logically entailed by premises.
- [] Resolution is refutation complete, meaning it can always detect unsatisfiability by deriving an empty clause.
- [] Proof by contradiction with resolution involves negating the goal, adding it to the knowledge base, and deriving an empty clause.
- [] An empty clause signifies a contradiction, proving the original goal to be true.
- [] Resolution is more general than Modus Ponens and can perform all inferences that Modus Ponens can.

Conclusion

Resolution proof systems provide a robust and systematic method for logical inference in Artificial Intelligence. By standardizing knowledge representation into Conjunctive Normal Form and leveraging the principle of proof by contradiction, AI agents can effectively deduce new information and validate hypotheses. This capability is not merely theoretical; it underpins the reasoning abilities of expert systems, automated theorem provers, and various decision-making AI applications.

The refutation completeness of resolution, while not a universal completeness, is precisely what makes it so valuable for AI. It guarantees that if a logical inconsistency exists within a knowledge base and a negated query, resolution will uncover it, thereby confirming the truth of the original query. This fundamental mechanism allows AI systems to move beyond simple pattern matching to engage in deeper, more verifiable forms of reasoning, forming a critical component in the ongoing quest to build truly intelligent machines.